CURRICULUM VITAE SHIWEI LIU

SHIWEI LIU

Web: Google Scholar Personal Website
Contact: s.liu3@tue.nl

EDUCATION

Eindhoven University of Technology (TU/e), The Netherlands *Mar 2018 - April 2022*
Ph.D., Department of Mathematics and Computer Science
w. Cum Laude/Distinguished Dissertations
Promotors: Prof. Dr. Mykola Pechenizkiy; Dr. Decebal Constantin Mocanu

Harbin Institute of Technology, China *Sep 2015 - Jul 2017*
M.Eng, School of Mechanical Engineering and Automation

North University of China, China *Sep 2011 - Jul 2015*
B.Eng., Control Engineering

PROFESSIONAL EXPERIENCE

Newton International Fellow *Jan 2024 - Present*
University of Oxford, UK
Mathematical Institute

Postdoctoral Fellow *Sep 2022 - Oct 2023*
University of Texas at Austin (UT Austin), USA
Institute for Foundations of Machine Learning (IFML)
Host: Dr. Zhangyang (Atlas) Wang

Doctoral Researcher *Mar 2018 - Apr 2022*
Eindhoven University of Technology, the Netherlands
Supervisors: Prof. Dr. Mykola Pechenizkiy, Dr. Decebal Constantin Mocanu

Research Intern *Jun 2021 - Nov 2021*
JD Explore Academy, China

Research Intern *Mar 2017 - May 2017*
CUHK-Shenzhen, China

RESEARCH INTERESTS

- **Machine Learning (ML):** Learning with Sparsity, Efficient Training and Inference, Transformers (esp. LLMs), ML Robustness, Privacy and Fairness.

AWARDS AND HONOURS

- Rising Star Award, Conference on Parsimony and Learning (CPAL), Link *10/2023*
- Best Ph.D. Dissertation Award Runner-Up, Informatics Europe, Link *10/2023*
- Newton International Fellowship Award, Royal Society & British Academy, 8%, Link *09/2023*
- Carnegie Bosch Fellowship, Carnegie Mellon University & Bosch, Link *04/2023*
- Best Paper Award, Learning on Graphs Conference (LoG 2022) *11/2022*
- AAAI-23 Travel Scholarship, USA *12/2022*
- Cum Laude (Distinguished Ph.D. thesis), Eindhoven University of Technology, NL, 5% *4/2022*

- Outstanding Intern, JD Explore Academy, 2% *10/2021*
- The First Class Scholarship, Harbin Institute of Technology, China *07/2017*
- China National Scholarship, China, 0.2% *10/2016*
- The First Class Scholarship, Harbin Institute of Technology, China *07/2016*
- The Outstanding Graduate, North University of China, China *07/2015*

GRANTS (TOTAL: $1,112,177)

- Shiwei Liu. (Single Lead Investigator), "Newton International Fellowship Award", Sponsored by Royal Society & British Academy, £394,851 ($481,746). Expected January 2024 - January 2027. *Success rate: 8%.*
- Shiwei Liu. (Lead Investigator), "Efficient Neural Network Scaling through Sparsity", Sponsored by Dutch Research Council: NWO Grants for Computing Time on the National Computer Facilities, $90,431.5. Expected November 2023 - November 2025.
- Shiwei Liu. (Single Lead Investigator), "Carnegie Bosch Fellowships", Sponsored by CMU and Bosch, $160,000. 2023 - 2025 *Success rate: 10%. (declined)*
- Shiwei Liu. (Lead Investigator), "Large-Scale Sparse Neural Models.", Sponsored by Dutch Research Council: NWO Grants for Computing Time on the National Computer Facilities, $220,000. Expected April 2022 - April 2024
- Shiwei Liu. (Single Lead Investigator), "Postdoctoral Fellowship", Sponsored BY The NSF AI Institute for Foundations of Machine Learning (IFML), and the NSF TRIPODS, $160,000. Expected September 2022 - September 2024

INVITED TALKS

- University of Edinburgh. Organizer: Sotirios Tsaftaris (University of Edinburgh). Jul. 2023
- Zhidx. Organizer: Zhidx Jul. 2023
- CTSTA Workshop in PLDI 2023. Organizer: Fredrik Kjolstad (Stanford), Saman Amarasinghe (MIT), Michelle Mills Strout (University of Arizona) Jun. 2023
- EfficientML Reading Group. Organizer: Olga Saukh (TU Graz) May. 2023
- Sparsity Reading Group. Organizer: Anna Golubeva (MIT), Dan Alistarh (IST Austria), Decebal Constantin Mocanu (University of Luxembourg), Gintare Karolina Dziugaite (Google Brain, McGill University), Utku Evci (Google Brain), Yani Ioannou (University of Calgary) Mar. 2023
- Sharc Group, Organizer: Cong Hao (Georgia Institute of Technology) Mar. 2023
- IFML, Organizer: Adam Klivans (UT Austin), Alexandros Dimakis (UT Austin) Jan. 2023
- LoG2022 Meetup, Organizer: Jure Leskovec (Stanford), Kexin Huang (Stanford) Dec. 2022
- NCSU Reliable & Efficient Computing Lab, Organizer: Dongkuan Xu (NC State) Nov. 2022
- EIC Lab, Organizer: Yingyan Lin (Georgia Institute of Technology) Sep. 2022
- AI XinQingNian Nov. 2022
- Data Mining Group, Organizer: Mykola Pechenizkiy (TU Eindhoven) May. 2022

EDUCATIONAL AND TEACHING ACTIVITIES

Teaching

- 2IMM00 Seminar Data Mining and AI, Eindhoven University of Technology Oct. 2023
- HA001 Honors Academy Bachelor, Eindhoven University of Technology Nov. 2023

Tutorial Presentation

- ICASSP 2024, "Sparsity in Large Language Models: The New Odyssey", Seoul, Korea, Apr. 2024
Co-organizer & Presenter; 75-min tutorial presentation on sparsity in Large Language Models (LLMs).
- VITA group workshop, "Sparsity and Efficiency", Austin, USA Sep. 2023
Presenter; 60-min tutorial presentation on recent works on sparse and efficient LLMs.
- IJCAI 2023, "Sparse Training for Supervised, Unsupervised, Continual, and Deep Reinforcement Learning with Deep Neural Networks", Macao, China. Link Aug. 2023

Co-organizer & Presenter; one-hour tutorial presentation from sparse training to sparse scaling.
• ECMLPKDD 2022, "Sparse Neural Network Training", Grenoble, France. Link Sep. 2022
Co-organizer & Presenter; 45-min tutorial presentation on understanding sparse neural network training with supervised learning.

Students Mentoring

PhD Students - Research Advisor:

• Lu Yin, Eindhoven University of Technology Dec. 2021 - Current
• Tianjin Huang, Eindhoven University of Technology Dec. 2021 - Current
• Boqian Wu, University of Twente Aug. 2022 - Current
• Qiao Xiao, Eindhoven University of Technology Aug. 2021 - Current
• Ajay Kumar Jaiswal, University of Texas at Austin Sep. 2022 - Current
• Duc N.M Hoang, University of Texas at Austin Sep. 2022 - Current
• Zhenyu Zhang, University of Texas at Austin Sep. 2022 - Current

Master Students - Research Advisor:

• Tiansheng Huang, South China University of Technology Jun. 2021 - Dec. 2021

Undergraduate Students - Research Advisor:

• Honors Academy: Simon Sukup, Austin Roose, Angelos Mangos, Mikoaj Pujanek, Eleftheria Kolokytha, Eindhoven University of Technology Nov. 2023 - Present

SERVICE

2024: Conference Reviewer: ICLR, AAAI, AISTATS
2023: Area Chair: ICIP
Conference Reviewer: NeurIPS, ICML, CVPR, ICCV, AAAI, UAI, DAC
Journal Reviewer: JMLR, TPAMI
2022: Area Chair: ICIP
Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, AAAI
Journal Reviewer: TPAMI
2021: Conference Reviewer: NeurIPS, ICLR, ICML, AISTATS
Journal Reviewer: IEEE Transactions on Evolutionary Computation
2020: Conference Reviewer: ECMLPKDD, IDA
Journal Reviewer: ACM Transactions On Intelligent Systems And Technology

ORGANIZATIONAL CONTRIBUTION

Tutorial Organization

• ICASSP 2024, "Sparsity in Large Language Models: The New Odyssey", Seoul, Korea. Apr. 2024
• IJCAI 2023, "Sparse Training for Supervised, Unsupervised, Continual, and Deep Reinforcement Learning with Deep Neural Networks", Macao, China. Link Aug. 2023
• ECMLPKDD 2022, "Sparse Neural Network Training", Grenoble, France. Link Sep. 2022

Event Organization

• NeurIPS 2023 Local Meetup, Eindhoven University of Technology, the Netherlands. Dec. 2023
• ICLR 2022 Local Meetup, Eindhoven University of Technology, the Netherlands. Apr. 2022

MEDIA COVERAGE

• Press Release of Early Career Research Schemes of Royal Society in Establishing the Next Generation of Research Leaders in the UK. Royal Society. Oct. 2023. Link
• Best Dissertation Award Runner-up Award. Informatics Europe. Oct. 2023. Link
• CPAL Rising Star Award. Conference on Parsimony and Learning (CPAL). Oct. 2023. Link
• Shiwei Liu and VITA Group Receive Best Paper Award at LoG 2022. IFML News. Jan. 2023. Link

PEER REVIEWED PUBLICATIONS

Publication Summary

Overall: **43** papers (24 **A*** and 3 **A** conference paper, CORE Conference Ranking), **5** journal papers, and **8** preprints; among them **19** first-author papers and **5** last-author papers.
Impact: **608** citations, h-index: **12**, i10-index: **13** (as of November 10 2023).

Peer-Reviewed Conference Publications (Sorted by date)

[**1**] Lu Yin, Gen Li, Meng Fang, Li Shen, Tianjin Huang, Zhangyang Wang, Vlado Menkovski, Xiaolong Ma, Mykola Pechenizkiy, **Shiwei Liu**. Dynamic Sparsity Is Channel-Level Sparsity Learner. The Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023. & SNN Workshop at ICLR 2023 **[Spotlight]**.

[**2**] Jaiswal, Ajay, **Shiwei Liu**, Tianlong Chen, and Zhangyang Wang. *The Emergence of Essential Sparsity in Large Pre-trained Models: The Weights that Matter.* The Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023.

[**3**] Duc N.M Hoang, Souvik Kundu, **Shiwei Liu**, Zhangyang Wang. Dont just prune by magnitude! Your mask topology is a secret weapon. The Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023.

[**4**] Hoang Pham, The-Anh Ta, **Shiwei Liu**, Lichuan Xiang, Dung D. Le, Hongkai Wen, Long Tran-Thanh. *Towards Data-Agnostic Pruning At Initialization: What Makes a Good Sparse Mask?* The Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023.

[**5**] Enneng Yang, Li Shen, Zhenyi Wang, **Shiwei Liu**, Guibing Guo, Xingwei Wang. *Data Augmented Flatness-aware Gradient Projection for Continual Learning.* International Conference on Computer Vision (ICCV), 2023.

[**6**] Ruisi Cai, Xiaohan Chen, **Shiwei Liu**, Jayanth Srinivasa, Myungjin Lee, Ramana Rao Kompella, Zhangyang Wang. *Scaling Federated Learning under Data and Task Heterogeneity: A Pilot Study.* International Conference on Computer Vision (ICCV), MTLFL FedVision Workshop 2023.

[**7**] Tianjin Huang*, **Shiwei Liu***, Tianlong Chen, Meng Fang, Li Shen, Vlado Menkovski, Lu Yin, Yulong Pei, Mykola Pechenizkiy. *Enhancing Adversarial Training via Reweighting Optimization Trajectory.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2023.

[**8**] Jiaxu Zhao, Lu Yin, **Shiwei Liu**, Meng Fang, Mykola Pechenizkiy. *REST: Debiasing Deep Neural Networks through Reweighted Sparse Training.* The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2023.

[**9**] Ajay Kumar Jaiswal, **Shiwei Liu**, Tianlong Chen, Ying Ding, Zhangyang Wang. *Instant Soup: Cheap Pruning Ensembles in A Single Pass Can Draw Lottery Tickets from Large Models.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023. **[Oral]**.

[**10**] Tianjin Huang, Lu Yin, Zhenyu Zhang, Li Shen, Meng Fang, Mykola Pechenizkiy, Zhangyang Wang, **Shiwei Liu**. *Are Large Kernels Better Teachers than Transformers for ConvNets?.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023.

[**11**] Ajay Kumar Jaiswal, **Shiwei Liu**, Tianlong Chen, Ying Ding, Zhangyang Wang. *Graph Ladling: Embarrassingly Scalable and Efficient Training of Powerful GNNs via Data-Centric Model Averaging.* The Fortieth International Conference on Machine Learning (ICML), PMLR, 2023.

[**12**] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Xuxi Chen, Qiao Xiao, Boqian Wu, Mykola Pechenizkiy, Decebal Mocanu, and Zhangyang Wang. *More Convnets in the 2020s: Scaling Up Kernels Beyond 51×51 Using Sparsity.* International Conference on Learning Representations (ICLR), 2023.

[13] Duc N.M Hoang, **Shiwei Liu**, Radu Marculescu, Zhangyang Wang. *Revisiting Pruning at Initialization Through the Lens of Ramanujan Graph.* International Conference on Learning Representations (ICLR), 2023. **[Oral]**.

[14] **Shiwei Liu***, Tianlong Chen*, Zhenyu Zhang, Xuxi Chen, Tianjin Huang, Ajay Kumar Jaiswal, and Zhangyang Wang. *Sparsity May Cry: Let Us Fail (Current) Sparse Neural Networks Together!.* International Conference on Learning Representations (ICLR), 2023. **[Spotlight]**.

[15] Tianlong Chen, Zhenyu Zhang, Ajay Kumar Jaiswal, **Shiwei Liu**, Zhangyang Wang *Sparse MoE with Random Routing as the New Dropout: Training Bigger and Self-Scalable Models.* International Conference on Learning Representations (ICLR), 2023. **[Spotlight]**.

[16] **Shiwei Liu**, and Zhangyang Wang., 2023. *Ten Lessons We Have Learned in the New "Sparseland": A Short Handbook for Sparse Neural Network Researchers.* SNN Workshop at ICLR 2023. **[Spotlight]**.

[17] Lu Yin*, **Shiwei Liu***, Fang Meng, Tianjin Huang, Vlado Menkovski, Mykola Pechenizkiy. *Lottery Pools: Winning More by Interpolating Tickets without Increasing Training or Inference Cost.* Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), 2023.

[18] Tianjin Huang, Tianlong Chen, Meng Fang, Vlado Menkovski, Jiaxu Zhao, Lu Yin, Yulong Pei, Decebal Constantin Mocanu, Zhangyang Wang, Mykola Pechenizkiy, **Shiwei Liu**. *You Can Have Better Graph Neural Networks by Not Training Weights at All: Finding Untrained Graph Tickets.* Learning on Graphs Conference (LoG), 2022. **[Oral & Best Paper Award]**.

[19] Qiao Xiao, Boqian Wu, Yu Zhang, **Shiwei Liu**, Mykola Pechenizkiy, Elena Mocanu, Decebal Constantin Mocanu. *Dynamic Sparse Network for Time Series Classification: Learning What to See.* The Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022.

[20] Lu Yin, Vlado Menkovski, Meng Fang, Tianjin Huang, Yulong Pei, Mykola Pechenizkiy, **Shiwei Liu**. *Superposing Many Tickets into One: A Performance Booster for Sparse Neural Network Training.* The 38th Conference on Uncertainty in Artificial Intelligence (UAI), 2022.

[21] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Li Shen, Decebal Constantin Mocanu, Zhangyang Wang, and Mykola Pechenizkiy. *The Unreasonable Effectiveness of Random Pruning: Return of the Most Naive Baseline for Sparse Training.* International Conference on Learning Representations (ICLR), 2022. Paper Code

[22] **Shiwei Liu**, Tianlong Chen, Zahra Atashgahi, Xiaohan Chen, Ghada Sokar, Elena Mocanu, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Deep Ensembling with No Overhead for either Training or Testing: The All-Round Blessings of Dynamic Sparsity.* International Conference on Learning Representations (ICLR), 2022. Paper Code

[23] **Shiwei Liu**, Tianlong Chen, Xiaohan Chen, Zahra Atashgahi, Lu Yin, Huanyu Kou, Li Shen, Mykola Pechenizkiy, Zhangyang Wang, and Decebal Constantin Mocanu. *Sparse Training via Boosting Pruning Plasticity with Neuroregeneration.* The Thirty-fifth Conference on Neural Information Processing Systems (NeurIPS), 2021. Paper Code

[24] **Shiwei Liu**, Lu Yin, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *Do We Actually Need Dense Over-Parameterization? In-Time Over-Parameterization in Sparse Training.* The Thirty-eighth International Conference on Machine Learning (ICML), PMLR, 2021. Paper Code

[25] **Shiwei Liu**, Decebal Constantin Mocanu, Yulong Pei, and Mykola Pechenizkiy. *Selfish sparse RNN training.* The Thirty-eighth International Conference on Machine Learning (ICML), PMLR, 2021.

[26] **Shiwei Liu**, Tim Van der Lee, Anil Yaman, Zahra Atashgahi, Davide Ferraro, Ghada Sokar, Mykola Pechenizkiy, and Decebal Constantin Mocanu. *Topological Insights into Sparse Neural Networks.* The European Conference on Machine Learning and Principles and Practice of Knowledge

Discovery in Databases (ECML-PKDD), Ghent, Belgium, 2020.

[**27**] **Shiwei Liu**. *Learning Sparse Neural Networks for Better Generalization.* 29th International International Joint Conference on Artificial Intelligence (IJCAI), 2020. Doctoral Consortium.

[**28**] **Shiwei Liu**, Decebal Constantin Mocanu, and Mykola Pechenizkiy. *On improving deep learning generalization with adaptive sparse connectivity.* The Workshop of Understanding and Improving Generalization in Deep Learning, The Thirty-sixth International Conference on Machine Learning (ICML), PMLR, 2019.

[**29**] Yan, Fulong, **Shiwei Liu**, and Nicola Calabretta. *Network Performance Optimization with Real Time Traffic Prediction in Data Center Network.* 2020 European Conference on Optical Communications (ECOC). IEEE, 2020.

Peer-Reviewed Journal Publications

[**30**] **Shiwei Liu**, Yuesong Tian, Tianlong Chen, Li Shen. *Don't Be So Dense: Sparse-to-Sparse GAN Training Without Sacrificing Performance.* International Journal of Computer Vision (IJCV).

[**31**] Zahra Atashgahi, Xuhao Zhang, Neil Kichler, **Shiwei Liu**, Lu Yin, Mykola Pechenizkiy, Raymond Veldhuis, Decebal Constantin Mocanu. *Feature Selection with Neuron Evolution.*, TMLR. 2023.

[**32**] Zahra Atashgahi, Joost Pieterse, **Shiwei Liu**, Decebal Constantin Mocanu, Raymond Veldhuis, Mykola Pechenizkiy. *A Brain-inspired Algorithm for Training Highly Sparse Neural Networks.* Machine Learning Journal (ECML-PKDD 2022 journal track).

[**33**] **Shiwei Liu**, Decebal Constantin Mocanu, Amarsagar Reddy Ramapuram Matavalam, Yulong Pei, and Mykola Pechenizkiy. (2021). *Sparse evolutionary deep learning with over one million artificial neurons on commodity hardware.* Neural Computing and Applications, 1-16.

[**34**] **Shiwei Liu**, Iftitahu Nimah, Vlado Menkovski, Decebal Constantin Mocanu, and Mykola Pechenizkiy. (2021). *Efficient and effective training of sparse recurrent neural networks.* Neural Computing and Applications, 1-12.

Pre-print and Under-Review

[**35**] Lu Yin, You Wu, Zhenyu Zhang, Cheng-Yu Hsieh, Yaqing Wang, Yiling Jia, Mykola Pechenizkiy, Yi Liang, Zhangyang Wang, **Shiwei Liu.** *Outlier Weighed Layerwise Sparsity (OWL): A Missing Secret Sauce for Pruning LLMs to High Sparsity.* arXiv preprint arXiv:2310.05175 (2023). Under Review.

[**36**] Yuxin Zhang, Lirui Zhao, Mingbao Lin, Yunyun Sun, Yiwu Yao, Xingjia Han, Jared Tanner, **Shiwei Liu**, and Rongrong Ji. *Dynamic Sparse No Training: Training-Free Fine-tuning for Sparse LLMs.* arXiv preprint arXiv:2310.08915 (2023). Under Review.

[**37**] Boqian Wua, Qiao Xiaob, **Shiwei Liu**, Lu Yin, Mykola Pechenizkiy, Decebal Constantin Mocanu, Maurice Van Keulen, Elena Mocanu. *E2ENet: Dynamic Sparse Feature Fusion for Accurate and Efficient 3D Medical Image Segmentation.* Under Review.

[**38**] **Shiwei Liu**, Tianjin Huang, Tianlong Chen, Zhangyang Wang. *The Counterattack of CNNs in Self-Supervised Learning: Larger Kernel Size Might be All You Need.* Under Review.

[**39**] Tiansheng Huang, **Shiwei Liu**, Li Shen, Fengxiang He, Weiwei Lin, Dacheng Tao., 2022. *Achieving Personalized Federated Learning with Sparse Local Models. arXiv preprint arXiv:2201.11380.* Paper

[**40**] Haoyu Ma, Chengming Zhang, Lizhi Xiang, Xiaolong Ma, Geng Yuan, Wenkai Zhang, **Shiwei Liu**, Tianlong Chen, Dingwen Tao, Yanzhi Wang, Zhangyang Wang, Xiaohui Xie. *HRBP: Hardware-friendly Regrouping towards Block-wise Pruning for Sparse Training.* Under Review.

[**41**] Yin Nan, Li Shen, Mengzhu Wang, **Shiwei Liu**, Chong Chen, Xian-Sheng Hua, Xiao Luo. *SPORT: A Subgraph Perspective on Graph Classification with Label Noise.* Under Review.

[**42**] Zhangheng Li, **Shiwei Liu**, Tianlong Chen, Ajay Kumar Jaiswal, Zhenyu Zhang, Dilin Wang, Raghuraman Krishnamoorthi, Shiyu Chang, Zhangyang Wang. *Sparse Cocktail: Co-Training Many Sparsity Patterns and Ratios at Once.* Under Review.

Ph.D. Thesis

[**43**] **Shiwei Liu**. Sparse Neural Network Training with In-Time Over-Parameterization. The Eindhoven University of Technology. 2022. Paper